

Table of Contents

1	Introduction	2
2	Exploratory Analysis and Initial Model	2
2.1	Lasso Regression	2
2.2	Data cleaning	2
2.3	Visualisation	2
3	Modelling	4
3.1	Baseline Linear Regression Models	4
3.1.1	Pooled Models	4
3.1.2	Unpooled Models	4
3.1.3	Partial Pooling	5
3.2	Hierarchical Models	5
3.2.1	Single Level Model	5
3.2.2	Multilevel Model	6
4	Results	7
5	Conclusion	7
6	Interpretation for Lay Audience	8
7	Limitations and Improvements	8

1 Introduction

The aim of this report is to better understand the main factors that impact student performance in two Portuguese secondary schools. For that we are going to use "Student Performance" ¹ data-cleaned data set from the UC Irvine Machine Learning Repository which attributes include student grades, demographic, socioeconomic and other school related variables and will focus on predicting final Portuguese language grades as the data set also includes final math grades. The analysis will be done using the R programming language. This investigation could be part of the guidance for governments when implementing new policies to improve education levels in the most efficient way. In our analysis, "**G3**" will be the outcome variable, ranging from 1 to 20. Originally there are 33 variables in total, and we reduce that number to the 10 most relevant ones (all predictors are categorical), which are the following:

- **G3**: Final Portuguese language grade
- **school**: Student's school (either "GP" (Gabriel Pereira) or "MS" (Mousinho da Silveira))
- **studytime**: Weekly study time in hours (1: <2 hours, 2: 2 to 5 hours, 3: 5 to 10 hours, 4: >10 hours)
- **failures**: Number of past class fails
- **schoolsup**: Extra educational support (Yes or No)
- **higher**: Wants to go higher education (Yes or No)
- **Dalc**: Workday alcohol consumption (from 1: very low to 5: very high)
- **goout**: Going out with friends (from 1: very low to 5: very high)
- **health**: Current health condition (from 1: very bad to 5: very good)
- **Fjob**: Mother's job (teacher, healthcare, civil services, at home or other)

2 Exploratory Analysis and Initial Model

2.1 Lasso Regression

Our original data set contains 33 variables which the number we want to reduce substantially in order to simplify and better understand the model. To do that, let's use the Lasso regression technique that shrinks the coefficients towards 0 and therefore will help to determine the coefficients we should exclude to avoid overfitting data. Mind that we won't include mother's job variable ("Fjob") as we save it for the hierarchical model.

After some analysis using R, we end up with $\lambda = 0.035$ and conclude the 8 most important coefficients which were mentioned above: **school**, **studytime**, **failures**, **schoolsup**, **higher**, **Dalc**, **goout** and **health**.

To analyse the performance of our model after removing most of the predictors, we will run a linear regression with all the predictors, and then one with only the most important variables. In particular, we will closely examine the R^2 coefficient, which measures how well the independent variables explain the variability of the dependent variable. That value drops from 0.444 to 0.3602 which is not too significant. The trade-off of lower R^2 is a simpler model with less coefficients to interpret.

2.2 Data cleaning

We will utilise correlation tests of "G3" against "G2" and "G1" which represent the second and first period grades respectively. Both tests calculate a really high correlation of 0.918548 and 0.8263871 respectively. Therefore we remove these predictors from our model. There are no missing values in the data set. We also split the data into training and testing with 90/10 ratio.

2.3 Visualisation

We will create a correlation matrix with all the selected variables to identify relationships between the predictors and the final Portuguese language grades, and any other multicollinearity issues. That correlation matrix is visualised below:

¹<https://archive.ics.uci.edu/dataset/320/student+performance>

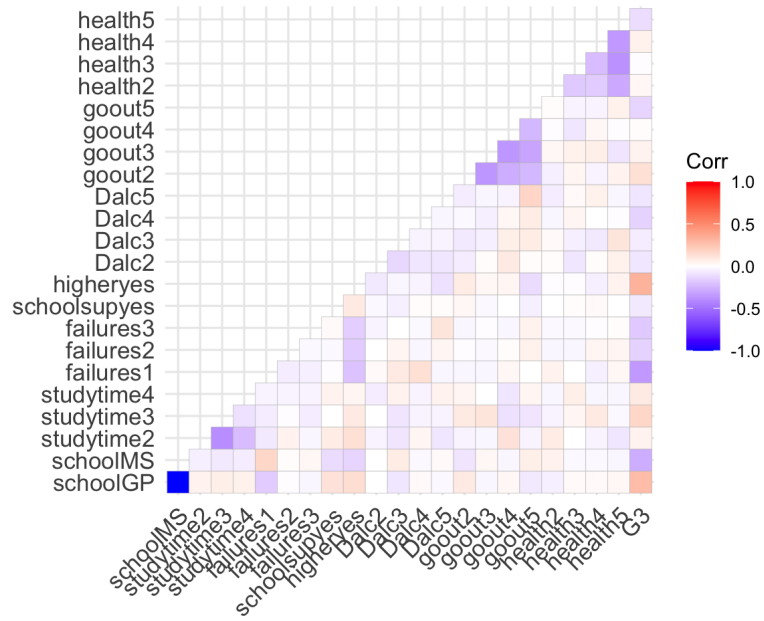


Figure 1: Correlation matrix

For the G3 column we detect small correlations between the outcome and predicted categories which is expected. Based on this figure, the biggest indicators of the final grade are “Dalc” and “failures” variables. No multicollinearity issues between variables is noticed.

For further exploratory analysis of the variables, we will make use of the boxplots which are visualised below. Note that no particular trends are observed.

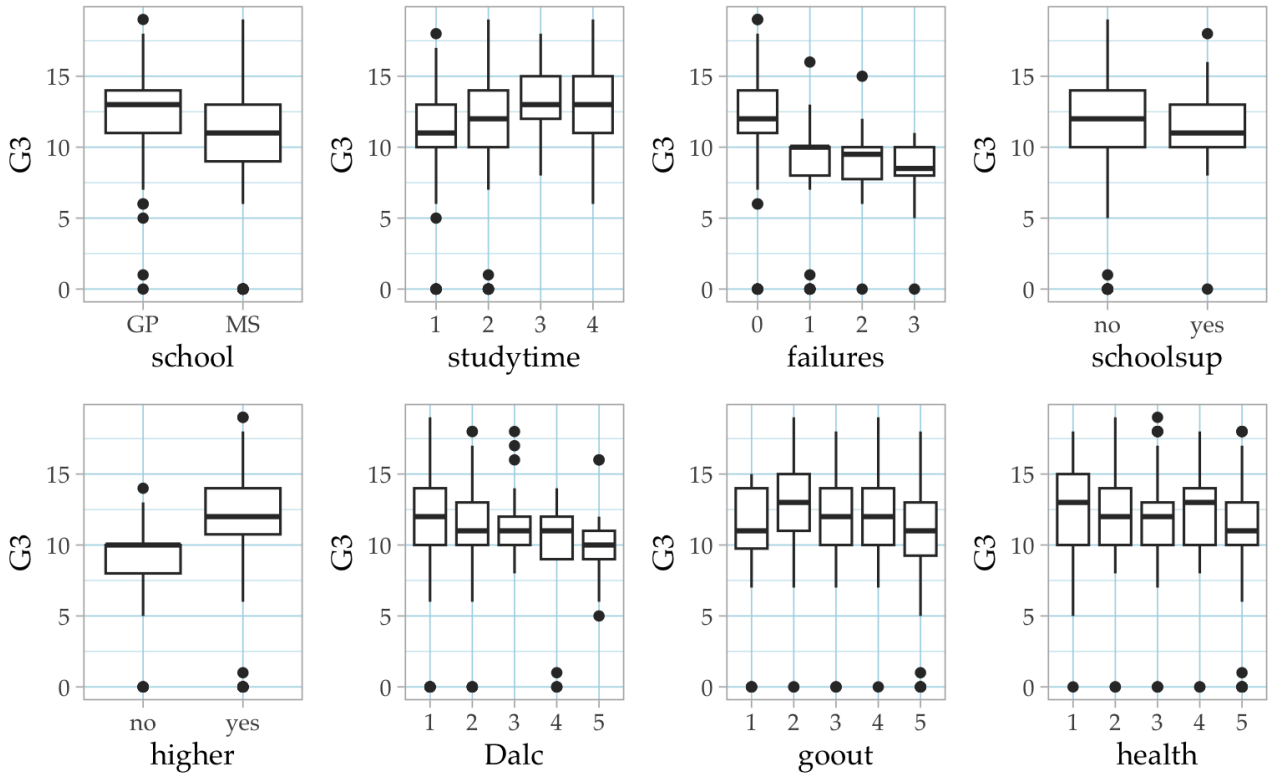


Figure 2: Boxplots of the associated predictors with the outcome G3

3 Modelling

3.1 Baseline Linear Regression Models

3.1.1 Pooled Models

The first model we will fit is the Linear Regression using all our important predictors, including the mother's job variable ("Fjob") as a dummy variable for each occupation which lets us obtain higher R^2 value.

We find the following estimated coefficients:

Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.80705	0.73769	14.650	< 2e-16 ***
schoolMS	-1.23107	0.23848	-5.162	3.40e-07 ***
studytime2	0.61232	0.26400	2.319	0.020736 *
studytime3	1.32142	0.35666	3.705	0.000232 ***
studytime4	1.44593	0.51191	2.825	0.004904 **
failures1	-2.77222	0.37824	-7.329	8.16e-13 ***
failures2	-2.68587	0.73356	-3.661	0.000275 ***
failures3	-3.01194	0.73039	-4.124	4.30e-05 ***
schoolsupyes	-1.34886	0.36580	-3.687	0.000249 ***
higheryes	1.76739	0.38898	4.544	6.78e-06 ***
Dalc2	-0.71130	0.29449	-2.415	0.016038 *
Dalc3	0.08780	0.45267	0.194	0.846274 .
Dalc4	-1.19811	0.72432	-1.654	0.098664 .
Dalc5	-1.23866	0.76458	-1.620	0.105783 .
goout2	1.52438	0.45740	3.333	0.000917 ***
goout3	0.99624	0.44220	2.253	0.024651 *
goout4	1.00159	0.46736	2.143	0.032540 *
goout5	0.41498	0.48397	0.857	0.391564 .
health2	-0.23334	0.43932	-0.531	0.595528 .
health3	-0.86854	0.39311	-2.209	0.027553 *
health4	-0.23976	0.40606	-0.590	0.555121 .
health5	-0.95926	0.35795	-2.680	0.007582 **
Fjobhealth	-0.56289	0.73894	-0.762	0.446533 .
Fjobother	-0.09674	0.45488	-0.213	0.831666 .
Fjobservices	-0.44202	0.47727	-0.926	0.354775 .
Fjobteacher	1.19042	0.64167	1.855	0.064098 .

Figure 3: Coefficients for the pooled linear regression model ($R^2 = 0.3724$)

3.1.2 Unpooled Models

Next, we will again fit a Linear Regression, however this time we won't use "Fjob" as a dummy variable. We will rather fit multiple Linear Regression models, each for different occupation of "Fjob" ending up with 5 different values for each coefficient. We have high standard errors which is expected as we don't have that many observation for each occupation.

Our estimated coefficients are summarised like this:

Variable <chr>	Average_Coefficient <dbl>	Average_Standard_Error <dbl>
(Intercept)	10.9365547	4.1978320
schoolMS	1.1930372	3.4979318
studytime2	4.1396344	2.9982653
studytime3	4.3234159	2.2981885
studytime4	5.9223619	3.8438219
failures1	-0.8369029	2.5649741
failures2	-3.0627365	1.4359729
failures3	-5.6995274	1.5639561
schoolsupyes	-0.4881232	1.9971794
higheryes	3.6995856	1.5641199
Dalc2	0.7493615	1.3653793
Dalc3	4.4653799	6.1680375
goout2	1.2454906	1.0068859
goout3	-4.5822074	5.9476125
goout4	-3.1074556	4.3793371
goout5	-1.8459657	3.7434244
health2	-0.4128001	1.4857042
health3	-5.3776425	5.5745243
health4	-3.3872243	2.3219347
health5	-6.5008289	5.8742427
Dalc5	-0.9131061	0.9641864
Dalc4	-1.7170923	1.0517246

Figure 4: Average means and standard errors for every variable over all unique linear regressions

3.1.3 Partial Pooling

To reduce to the variance of our coefficients we could use the partial pooling model (take the average between the coefficients of all the regressions and the population mean) which pulls more extreme estimates towards an overall average. This method could help us reduce the overfitting effect on our training data and therefore improve predictions.

3.2 Hierarchical Models

Our last models will use Hierarchical Bayesian Regression techniques making use of the “**Rstan**” package in R. We aim to understand mother’s occupation specific variations in our coefficients through Bayesian modeling, capturing both within-occupation and between-occupation differences. To achieve this, we will implement Hierarchical models at **one** and **two levels**.

3.2.1 Single Level Model

For the single level model we will assign a prior distribution (Normal Distribution in this case) to each coefficient and then run a regression using the MCMC (Markov Chain Monte Carlo) algorithm. The parameters we choose for our priors are such that they are generally weakly informative (larger variance of 1 or 2). More specifically they are centered around 0 to avoid overfitting the data as much as possible.

We have the following results:

Estimates:					
	mean	sd	10%	50%	90%
(Intercept)	10.950	0.535	10.277	10.942	11.627
schoolMS	-1.331	0.228	-1.620	-1.332	-1.038
studytime2	0.416	0.243	0.104	0.413	0.733
studytime3	1.152	0.318	0.737	1.158	1.557
studytime4	1.092	0.468	0.494	1.090	1.686
failures1	-2.625	0.359	-3.099	-2.622	-2.175
failures2	-1.922	0.563	-2.643	-1.914	-1.187
failures3	-2.066	0.590	-2.799	-2.068	-1.311
schoolsupyes	-1.125	0.334	-1.558	-1.126	-0.684
higheryes	1.767	0.325	1.357	1.770	2.182
Dalc2	-0.549	0.267	-0.895	-0.546	-0.202
Dalc3	-0.073	0.423	-0.621	-0.067	0.461
Dalc4	-1.601	0.570	-2.329	-1.609	-0.875
Dalc5	-0.619	0.563	-1.350	-0.610	0.089
goout2	0.951	0.384	0.452	0.951	1.443
goout3	0.609	0.373	0.135	0.610	1.090
goout4	0.600	0.380	0.115	0.596	1.078
goout5	-0.037	0.410	-0.553	-0.034	0.492
health2	-0.009	0.402	-0.541	-0.007	0.493
health3	-0.689	0.357	-1.141	-0.689	-0.238
health4	-0.165	0.371	-0.642	-0.166	0.308
health5	-0.857	0.316	-1.260	-0.867	-0.444
sigma	2.651	0.076	2.555	2.650	2.748

Figure 5: Estimates of the Single Level Model

3.2.2 Multilevel Model

For our multilevel model, we will assign a prior distribution to each coefficient, following with a hyperprior to each of these priors. Named distributions are Normal for the means and Cauchy for the variances. To ensure convergence of the posteriors, we use lower variances for the hyperpriors.

$$Intercept \stackrel{d}{\sim} Normal(\mu_a, \sigma_a)$$

$$\mu_a \stackrel{d}{\sim} Normal(10, 2)$$

$$\sigma_a \stackrel{d}{\sim} Cauchy(0, 3)$$

$$Intercept \stackrel{d}{\sim} Normal(\mu_b, \sigma_b)$$

$$\mu_a \stackrel{d}{\sim} Normal(0, 0.5)$$

$$\sigma_a \stackrel{d}{\sim} Cauchy(0, 1)$$

With 9000 iterations, no divergence transitions were detected. For diagnostics, we used Rhat and other convergence tools which showed sensible results. Therefore we end up with our final model:

Estimates:					
	mean	sd	10%	50%	90%
(Intercept)	9.576	0.516	9.027	9.561	10.177
schoolMS	0.071	0.129	-0.094	0.071	0.236
studytime2	0.591	0.125	0.430	0.591	0.751
studytime3	0.774	0.135	0.599	0.775	0.949
studytime4	0.663	0.141	0.480	0.665	0.844
failures1	0.150	0.138	-0.026	0.148	0.327
failures2	0.486	0.140	0.305	0.486	0.664
failures3	0.492	0.142	0.308	0.492	0.676
schoolsupyes	0.424	0.134	0.253	0.425	0.597
higheryes	0.902	0.137	0.728	0.902	1.077
Dalc2	0.409	0.132	0.241	0.409	0.576
Dalc3	0.520	0.139	0.340	0.521	0.699
Dalc4	0.485	0.144	0.303	0.485	0.670
Dalc5	0.535	0.141	0.353	0.536	0.714
goout2	0.762	0.132	0.592	0.762	0.932
goout3	0.653	0.128	0.486	0.655	0.815
goout4	0.595	0.131	0.430	0.593	0.765
goout5	0.415	0.135	0.241	0.415	0.589
health2	0.613	0.135	0.438	0.614	0.785
health3	0.550	0.131	0.382	0.550	0.719
health4	0.634	0.133	0.463	0.633	0.804
health5	0.384	0.129	0.217	0.386	0.550
b[(Intercept) Fjob:at_home]	-0.347	0.579	-1.072	-0.280	0.240
b[(Intercept) Fjob:health]	0.137	0.608	-0.541	0.089	0.845
b[(Intercept) Fjob:other]	-0.140	0.496	-0.716	-0.113	0.363
b[(Intercept) Fjob:services]	-0.389	0.521	-1.018	-0.352	0.119
b[(Intercept) Fjob:teacher]	0.845	0.676	0.059	0.766	1.711
sigma	3.172	0.094	3.053	3.170	3.294
Sigma[Fjob:(Intercept),(Intercept)]	1.059	2.146	0.055	0.498	2.318

Figure 6: Coefficients for the 2-level hierarchical model

4 Results

We have finally obtained all models and are ready to compare their performances using the test data. To do that, we will use MSE (Mean Squared Error) which measures the average squared difference between predicted and actual values in a regression model.

Model	MSE
Normal Regression	8.294633
Pooled Regression	8.605057
Unpooled Regression	65.899334
Hierarchical 1 Level	21.928399
Hierarchical 2 Levels	21.991764

Table 1: Mean Squared Error for all regressions using the test dataset

5 Conclusion

In this report we have investigated the relationship between students' final Portuguese language scores and their habits, socioeconomic factors etc. across 2 different Portuguese schools employing five different regression models and evaluating their performance on unseen data. We observe how the unpooled regression model performed by far the worst due to large variances leading to imprecise coefficients. The pooled regression model on the other hand has a quite similar MSE as the normal regression with a slightly higher R^2 value. This is a good performance considering the pooled model is much simpler. Both hierarchical models performed worse than the normal and pooled regression.

6 Interpretation for Lay Audience

To illustrate our findings regarding the student's final Portuguese language grade, let's use two hypothetical individuals who have some different factors that are relevant. For that we using the results of the pooled regression model due to its simplicity and good diagnostics. First, we have:

- **Person X** goes to Mousinho da Silveira, dedicates 5 to 10 hours every week to studying, has never failed any classes and is motivated to have higher education.
- **Person Y** goes to Gabriel Pereira, studies less than 2 hours weekly, has failed once and doesn't want to go higher education.

If we assume that Person X and Person Y are identical in all other ways except those above, we expect Person X to score 1.57 times higher in their final Portuguese language test than Person Y.

According to our model, students in Mousinho da Silveira score 11% lower than their peers in Gabriel Pereira which could point towards some difference in teaching quality levels between 2 schools.

We have learned that motivation to go into higher education plays a significant role in final Portuguese language scores. More specifically, students who want to have higher education, they have a 16% higher grade compared to those who don't.

Similarly students who dedicate 5 to 10 hours every week to studying, score 12% higher compared to their peers who dedicate less than 2. This result is rather intuitive.

Finally we have found that Person Y having 1 class fail leads to 25% reduction in their final Portuguese language grade compared to Person X's who hasn't failed any classes.

7 Limitations and Improvements

One main improvement our models could benefit from is a larger data size which would lower variances of our results and reduce measurement errors leading to more accurate results.